Gyeongdeok Seo

Ph.D. Student in Statistics

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Education

Ph.D. **University of Illinois Urbana-Champaign**, Statistics — Illinois, USA, Aug 2026 – present

M.S. **Yonsei University**, Statistics — Seoul, South Korea, Mar 2024 – Feb 2026
- GPA: 4.15/4.30.

B.A. **Yonsei University**, Applied Statistics — Seoul, South Korea, Mar 2016 – Aug 2022
- GPA: 3.97/4.30.
- Exchange student, Georgia Institute of Technology (Fall 2021).
- Leave of absence for military service, Republic of Korea Air Force (February 2018–January 2020).

Publications

Dissecting Causal Mechanism Shifts via FANS: Function And Noise Separation. Jan 2026
Gyeongdeok Seo, Jaeyoon Shim, Mingyu Kim, Hoyoon Byun, Yonghan Jung, Kyungwoo Song.
Gyeongdeok Seo, Jaeyoon Shim, Mingyu Kim, Hoyoon Byun, Yonghan Jung, Kyungwoo Song
proceedings.mlr.press/v306/seo26d.html (ICML 2026)

Robust Adaptation of Foundation Models with Black-Box Visual Prompting.
Changdae Oh, **Gyeongdeok Seo**, Geunyoung Jung, Zhi-Qi Cheng, Hosik Choi, Jiyoung Jung, Kyungwoo Song.
Changdae Oh, Gyeongdeok Seo, Geunyoung Jung, Zhi-Qi Cheng, Hosik Choi, Jiyoung Jung, Kyungwoo Song
arxiv.org/abs/2407.17491 (IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI))

Static and Dynamic Scoring Systems for Post-acute Sequelae of SARS-CoV-2 in a Korean Cohort.
Gyeongdeok Seo*, Hyejin Joo*, Kyungwoo Song*, Mingyu Kim, et al., Jaehun Jung, Jacob Lee. *Equal contribution.*
Gyeongdeok Seo*, Hyejin Joo*, Kyungwoo Song*, Mingyu Kim, et al., Jaehun Jung, Jacob Lee
www.ijidonline.com/article/S1201-9712(26)00013-5/fulltext (International Journal of Infectious Diseases)

CCL: Causal-aware In-context Learning for Out-of-Distribution Generalization. Jan 2025
Hoyoon Byun, **Gyeongdeok Seo**, Joonseong Kang, Taero Kim, Jihee Kim, Kyungwoo Song.
Hoyoon Byun, Gyeongdeok Seo, Joonseong Kang, Taero Kim, Jihee Kim, Kyungwoo Song
neurips.cc/virtual/2025/loc/san-diego/poster/119001 (NeurIPS 2025)

COVID-19 Prediction with Doubly Multi-task Gaussian Process. Jan 2025
Sooyon Kim, Yongtaek Lim, Sungjun Lim, **Gyeongdeok Seo**, Jihee Kim, Hojun Park, Jaehun Jung, Kyungwoo Song.
Sooyon Kim, Yongtaek Lim, Sungjun Lim, Gyeongdeok Seo, Jihee Kim, Hojun Park, Jaehun Jung, Kyungwoo Song
www.sciencedirect.com/science/article/pii/S1532046425001017 (Journal of Biomedical Informatics (2025))

First Step for Theoretical and Practical Foundations of Robust Visual Prompting. Jan 2024
Gyeongdeok Seo*, Changdae Oh*, Kyungwoo Song. *Equal contribution.*
Gyeongdeok Seo*, Changdae Oh*, Kyungwoo Song
trustaiijcai.github.io/trustai-ijcai24/papers (IJCAI 2024 Trustworthy AI Workshop)

Experience

Yonsei University, Graduate Researcher — advised by Prof. Kyungwoo Song — Mar 2024 – Dec 2025, 1 year 10 months
- Developed FANS (Function And Noise Separation), a framework using normalizing flows to detect and dissect causal mechanism shifts into changes in the causal function or the noise distribution.
- Proposed a statistical criterion based on independence between flow-generated noise residuals and parent variables to distinguish function-driven and noise-driven shifts in structural causal models.
- Filed a Korean patent application, *Method and Apparatus for Detecting and Analyzing Causal Mechanism Shifts* (pending).

Vaccine Safety and Causality Monitoring Project, Graduate Researcher — advised by Prof. Kyungwoo Song — Mar 2024 – Dec 2025, 1 year 10 months
Establishment of a monitoring system and methodology to verify the causality of adverse reactions of major interest after vaccination.
- Designed static and dynamic scoring models for post-acute sequelae of SARS-CoV-2 (PASC) in a Korean cohort using logistic lasso and functional logistic lasso regression; identified differences in chest pain scores among Omicron wave sub-cohorts.
- Constructed a dynamic scoring system with time-varying coefficients based on Functional Principal Component Analysis (FPCA) of longitudinal data.

Global Research on Regulatory Evaluation of Biohealth Products, Research Project Participant — Apr 2024 – Feb 2026, 1 year 11 months
Global Collaborative Research for Advancing Regulatory Science and Technology in Safety and Efficacy Evaluation of Biohealth Products, funded by the Ministry of Food and Drug Safety, Korea.
- Presented chapters of *Causal Inference: What If* in a bi-weekly seminar and presented on detecting mechanism shifts to identify unmeasured confounders.

NAVER FINANCIAL, Data Analyst — Seoul, South Korea, Oct 2022 – Feb 2024, 1 year 5 months
- Engineered a data mart and time-series forecasting models to automate reporting; developed interactive controllers for a commerce expert and integrated outputs into dashboards for proactive risk detection.
- Analyzed Naver Pay payment trends and simulated promotion scenarios to support business decisions.

CC&I Research, Student Research Assistant (work-study) — Seoul, South Korea, Mar 2020 – Feb 2021, 1 year
- Evaluated non-inferiority of a client's new device against the original using the Wilcoxon signed-rank test to compare study endpoints.

Awards

Yonsei Blue Butterfly Scholarship
Full tuition for undergraduate studies.

Yongwoon Scholarship
Full tuition for master's studies.

NRF Research Encouragement Scholarship
Research encouragement scholarship for master's students.

Award by the Chief of Maekyung Media Group Dec 2021
2021 NIA Big Data Contest.

Award by the Chief of Korea Land & Housing Corporation Aug 2021
2021 DACON.

Award by NH Investment & Securities Feb 2021
2021 DACON.

Teaching

Yonsei University — Teaching Assistant: STA1001 Introduction to Statistics, June–July 2024.

Skills

Programming
Advanced: Python, R, SQL

Interests

Research interests
Causal Inference, Structural Causal Models, Probabilistic Machine Learning